



TIME SERIES ANALYSIS

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ARMA processes

- Combine AR and MA processes
- An ARMA process of order (p,q) is given by

$$y_t = \alpha_1 y_{t-1} + \cdots + \alpha_p y_{t-p} + e_t + \beta_1 e_{t-1} + \cdots + \beta_q e_{t-q}$$



ARMA processes

- Alternative expressions are possible using the backshift operator

$$\varphi(B)y_t = \theta(B)e_t$$

Where

$$\varphi(B) = 1 + \alpha_1 B + \cdots + \alpha_p B^p$$

$$\theta(B) = 1 + \beta_1 B + \cdots + \beta_q B^q$$



ARMA processes

- A random process that contains both auto-regressive and moving average parts is said to be an ARMA(L_{AR} , L_{MA}) process.

$$v[k] = \frac{C(q)}{D(q)} e[k] = \frac{1 + \sum_{n=1}^{L_{MA}} c_n q^{-n}}{1 + \sum_{n=1}^{L_{AR}} d_n q^{-n}} e[k]$$



Summary of the Behavior of autocorrelation and partial autocorrelation functions

Behavior of autocorrelation and partial autocorrelation functions

Model	AC	PAC
Autoregressive of order p $y_t = \delta + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + e_t$	Dies down	Cuts off after lag p
Moving Average of order q $y_t = \delta + e_t - \theta_1 e_{t-1} - \theta_2 e_{t-2} - \dots - \theta_q e_{t-q}$	Cuts off after lag q	Dies down
Mixed Autoregressive-Moving Average of order (p,q) $y_t = \delta + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + e_t - \theta_1 e_{t-1} - \theta_2 e_{t-2} - \dots - \theta_q e_{t-q}$	Dies down	Dies down



Mixed Autoregressive-Moving Average (ARMA) Process of order (p, q)

□ Denoted by $ARMA(p, q)$

□ Contains p autoregressive and q moving average terms

□ The process is defined as

$$y_t = \delta + \phi_1 y_{t-1} + \phi_2 y_{t-2} \cdots + \phi_p y_{t-p} \\ + u_t + \theta_1 u_{t-1} + \theta_2 u_{t-2} \cdots + \theta_q u_{t-q} \quad (1)$$

$$E(y_t) = \mu \quad \forall t.$$



Taking expectation of (1), we have

$$\mu = \phi_1\mu + \phi_2\mu + \phi_3\mu + \cdots + \phi_p\mu + \delta$$

$$\Rightarrow \mu = \frac{\delta}{1 - \phi_1 - \phi_2 - \phi_3 - \cdots - \phi_p}$$

For $\delta = 0$, μ is also zero.

In terms of backward shift operator “B”:

$$\begin{aligned} & (1 - \phi_1 B - \phi_2 B^2 \cdots - \phi_p B^p) y_t \\ & = \delta + (1 + \theta_1 B + \theta_2 B^2 \cdots + \theta_q B^q) u_t \end{aligned} \quad (2)$$



Let

$$\Phi(B) \equiv (1 - \phi_1 B - \phi_2 B^2 \cdots - \phi_p B^p)$$

$$\Theta(B) \equiv (1 + \theta_1 B + \theta_2 B^2 \cdots + \theta_q B^q).$$

Then (2) can be represented as

$$\Phi(B)y_t = \delta + \Theta(B)u_t \tag{3}$$



Let

$$\Phi(B) \equiv (1 - \phi_1 B - \phi_2 B^2 \cdots - \phi_p B^p)$$

$$\Theta(B) \equiv (1 + \theta_1 B + \theta_2 B^2 \cdots + \theta_q B^q).$$

Then (2) can be represented as

$$\Phi(B)y_t = \delta + \Theta(B)u_t \tag{3}$$



ARMA Representation

- ☐ Approximates the dynamic of any purely nondeterministic weakly stationary process.
- ☐ Describes a weakly stationary process in terms of two polynomials, one for AR and the other for MA.
- ☐ Higher order AR or MA processes with large number of parameters can be approximated with lower order ARMA processes with lesser number of parameters.



$ARMA(1,1)$ Process:

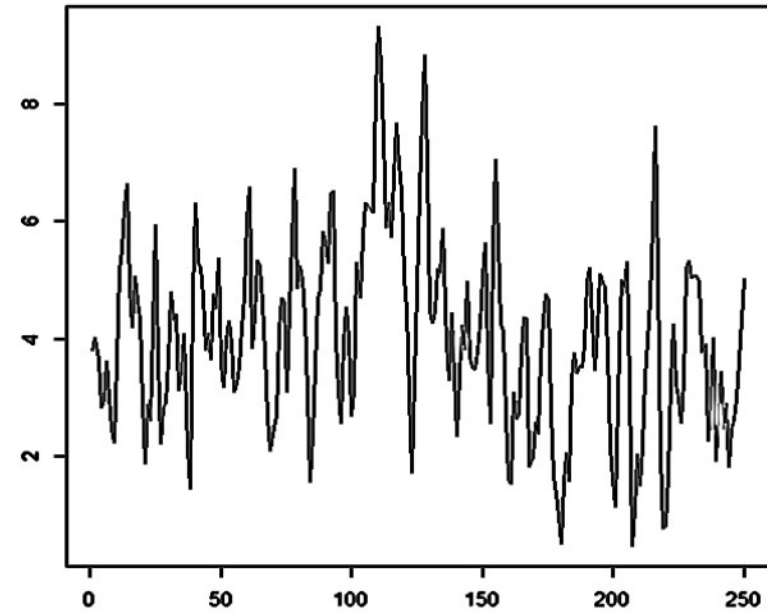
Consider $ARMA(1,1)$ process with $\delta=0$

$$y_t = \phi_1 y_{t-1} + u_t + \theta_1 u_{t-1} \quad (5)$$

The variance of the process is given by

$$\begin{aligned} \gamma_0 &= E(y_t^2) \\ &= E[\phi_1 y_{t-1} + u_t + \theta_1 u_{t-1}]^2 \\ &= E[\phi_1^2 y_{t-1}^2 + u_t^2 + \theta_1^2 u_{t-1}^2 + 2\phi_1 y_{t-1} u_t \\ &\quad + 2\phi_1 \theta_1 y_{t-1} u_{t-1} + 2\theta_1 u_t u_{t-1}] \end{aligned} \quad (6)$$

Simulated sample from
 $ARMA(1,1)$ process with
 $\phi_1 = 0.6, \theta_1 = 0.5$





Notice that

$$E[y_{t-1}u_t] = 0 = E[u_t u_{t-1}]$$

$$E[y_{t-1}u_{t-1}] = E[(\phi_1 y_{t-2} + u_{t-1} + \theta_1 u_{t-2})u_{t-1}] = \sigma_u^2.$$

Therefore, (6) reduces to

$$\gamma_0 = \phi_1^2 \gamma_0 + \sigma_u^2 + \theta_1^2 \sigma_u^2 + 2\theta_1 \phi_1 \sigma_u^2 \quad (7)$$

$$\Rightarrow \gamma_0 = \frac{\sigma_u^2(1 + \theta_1^2 + 2\theta_1 \phi_1)}{(1 - \phi_1^2)} = \sigma_y^2 \quad (8)$$



Further

$$\begin{aligned}\gamma_1 &= E[y_t y_{t-1}] \\ &= E[(\phi_1 y_{t-1} + u_t + \theta_1 u_{t-1}) y_{t-1}] \\ &= (\phi_1 \gamma_0 + \theta_1 \sigma_u^2)\end{aligned}$$

$$\Rightarrow \gamma_1 = \frac{\sigma_u^2(\phi_1 + \theta_1 + \phi_1 \theta_1^2 + \theta_1 \phi_1^2)}{(1 - \phi_1^2)}$$

$$\gamma_0 = \frac{\sigma_u^2(1 + \theta_1^2 + 2\theta_1 \phi_1)}{(1 - \phi_1^2)}$$

$$= \frac{\sigma_u^2(\phi_1 + \theta_1)(1 + \phi_1 \theta_1)}{(1 - \phi_1^2)}$$

(9)



$$\gamma_2 = E[y_t y_{t-2}]$$

$$= E[(\phi_1 y_{t-1} + u_t + \theta_1 u_{t-1}) y_{t-2}]$$

$$= \phi_1 \gamma_1 \quad (10)$$

$$\gamma_k = E[y_t y_{t-k}]$$

$$= E[(\phi_1 y_{t-1} + u_t + \theta_1 u_{t-1}) y_{t-k}]$$

$$= \phi_1 \gamma_{k-1}; \quad (k \geq 3) \quad (11)$$

$$= \phi_1^2 \gamma_{k-2}$$

$$= \phi_1^3 \gamma_{k-3}$$

$$E(y_{t-1}, y_{t-k})$$

$$= \gamma_{(t-1) - (t-k)}$$

$$= \gamma_{k-1}$$



ACF of ARMA(1,1) process:

$$\rho_1 = \frac{\gamma_1}{\gamma_0} = \frac{(\phi_1 + \theta_1)(1 + \phi_1\theta_1)}{(1 + \theta_1^2 + 2\phi_1\theta_1)} \quad (12)$$

For $k \geq 2$

$$\begin{aligned} \rho_k &= \frac{\gamma_k}{\gamma_0} \\ &= \phi_1 \frac{\gamma_{k-1}}{\gamma_0} = \phi_1 \rho_{k-1} \\ &= \phi_1^2 \rho_{k-2} = \cdots = \phi_1^{k-1} \rho_1 \end{aligned} \quad (13)$$



$$\rho_1 = \frac{(\phi_1 + \theta_1)(1 + \phi_1\theta_1)}{(1 + \theta_1^2 + 2\phi_1\theta_1)}; \rho_k = \phi_1^{k-1}\rho_1; k \geq 2$$

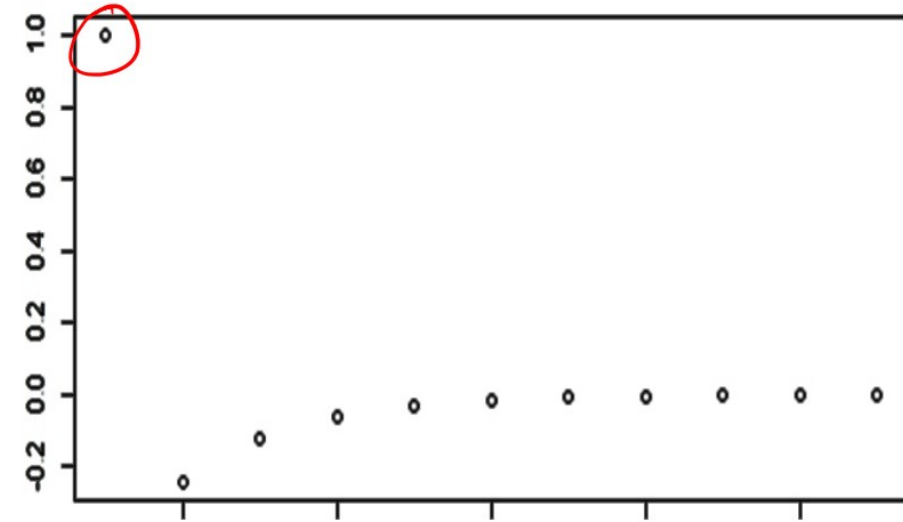
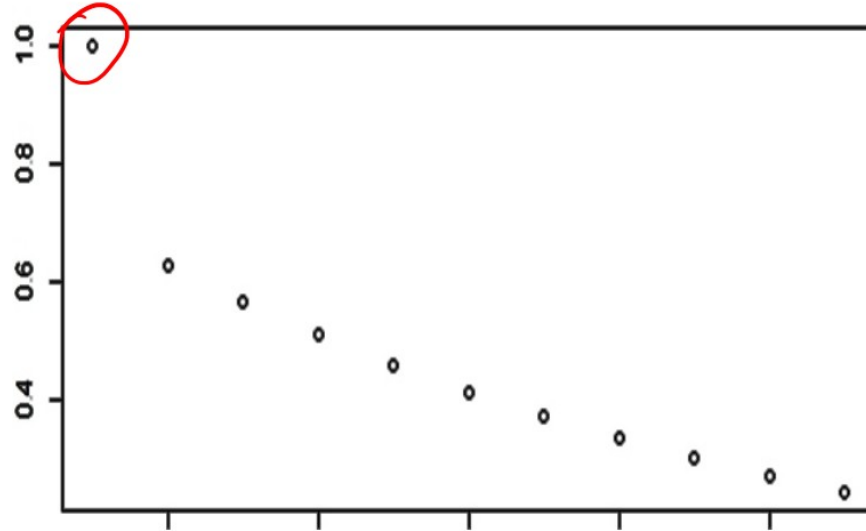
- i. If $\phi_1 + \theta_1 > 0, \phi_1 > 0, \rho_k > 0 \forall k$
 - ii. $\phi_1 + \theta_1 < 0, \phi_1 > 0, \rho_k < 0 \forall k$
 - iii. If $\phi_1 + \theta_1 > 0, \phi_1 < 0, \rho_k$ oscillates with $\rho_1 > 0$
 - iv. $\phi_1 + \theta_1 < 0, \phi_1 < 0, \rho_k$ oscillates with $\rho_1 < 0$
- $\forall k \geq 2, \rho_k$ decays exponentially in magnitude



ACF of ARMA(1,1) Process $y_t = \phi y_{t-1} + u_t + \theta u_{t-1}$

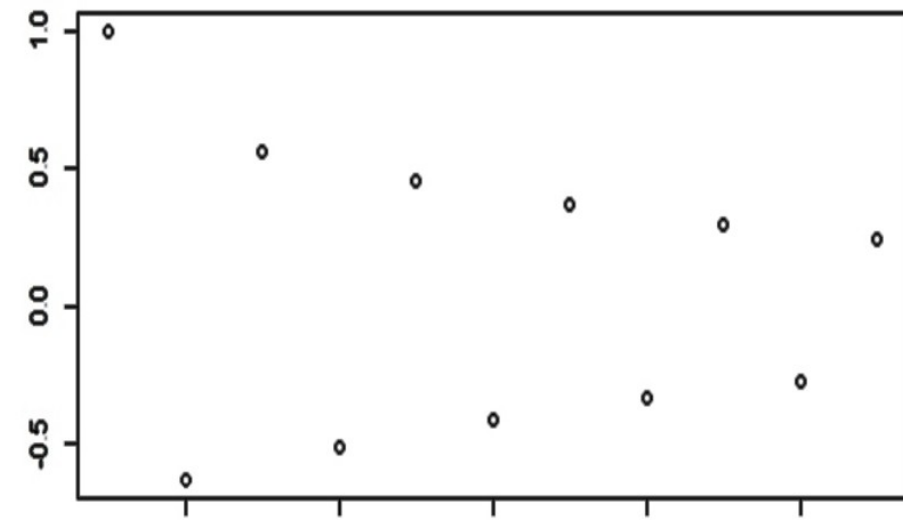
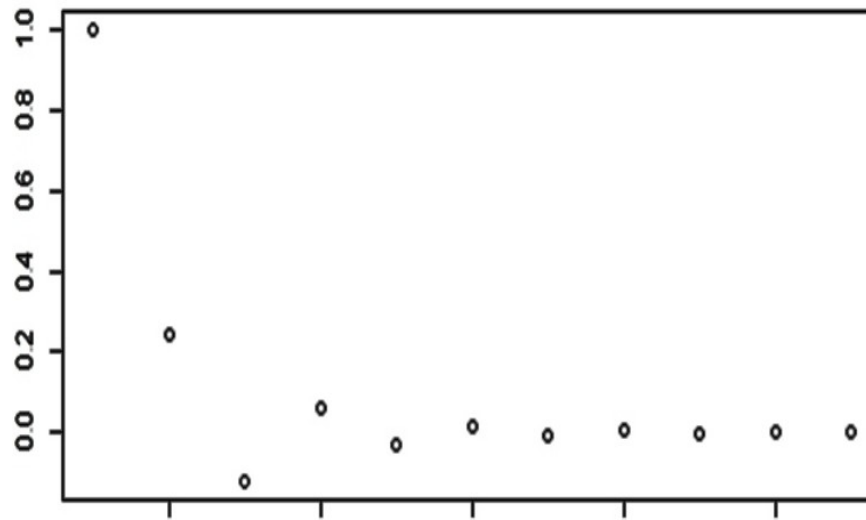
i. $\phi = 0.9, \theta = 0.5$

ii. $\phi = 0.5, \theta = 0.9$



iii. $\phi = -0.5, \theta = -0.9$

iv. $\phi = -0.9, \theta = -0.5$





ACVF and ACF of ARMA(p, q) Process:

$$\Phi(B)y_t = \Theta(B)u_t$$

If the process is stationary, we can write

$$y_t = \frac{\Theta(B)}{\Phi(B)} u_t = \underline{\Psi(B)u_t}$$

$$= \sum_{j=0}^{\infty} \psi_j B^j u_t = \sum_{j=0}^{\infty} \psi_j u_{t-j}$$



Variance:

$$\gamma_0 = \sigma_y^2 = E(y_t^2)$$

$$= E \left[y_t \{ (\phi_1 y_{t-1} + \phi_2 y_{t-2} \cdots + \phi_p y_{t-p}) \right. \\ \left. + (u_t + \theta_1 u_{t-1} + \theta_2 u_{t-2} \cdots + \theta_q u_{t-q}) \} \right]$$

$$= \phi_1 E(y_t y_{t-1}) + \phi_2 E(y_t y_{t-2}) + \cdots + \phi_p E(y_t y_{t-p}) \\ + E(y_t u_t) + \theta_1 E(y_t u_{t-1}) + \cdots + \theta_q E(y_t u_{t-q})$$

We have

$$E(y_t u_{t-l}) = E \left[\left(\sum_{j=0}^{\infty} \psi_j u_{t-j} \right) u_{t-l} \right] = \sigma_u^2 \psi_l \quad \forall l = 0, 1, \dots, q$$



$$E(y_t y_{t-l}) = E \left(\sum_{j=0}^{\infty} \psi_j u_{t-j} \sum_{j'=0}^{\infty} \psi_{j'} u_{t-j'-l} \right)$$

$$= E \left(\sum_{j=l}^{\infty} \psi_j \psi_{j-l} u_{t-j}^2 \right) = \sigma_u^2 \sum_{j=l}^{\infty} \psi_j \psi_{j-l}$$

$$\begin{aligned} E(y_t u_{t-l}) &= \sigma_u^2 \psi_l \\ \gamma_0 &= \sum_{l=1}^p \phi_l E(y_t y_{t-l}) \\ &\quad + E(y_t u_t) + \sum_{l=1}^q \theta_l E(y_t u_{t-l}) \end{aligned}$$

$$\text{Hence } \gamma_0 = \sigma_u^2 \left[\sum_{l=1}^p \sum_{j=l}^{\infty} \phi_l \psi_j \psi_{j-l} + \psi_0 + \sum_{l=1}^q \theta_l \psi_l \right]$$



ACVF:

$$\begin{aligned}\gamma_k &= E(y_t y_{t-k}) \\ &= E[(\phi_1 y_{t-1} + \phi_2 y_{t-2} \cdots + \phi_p y_{t-p} + u_t + \theta_1 u_{t-1} \\ &\quad + \theta_2 u_{t-2} \cdots + \theta_q u_{t-q}) y_{t-k}]\end{aligned}$$

For $1 \leq k \leq q$

$$\begin{aligned}\gamma_k &= \phi_1 \gamma_{k-1} + \phi_2 \gamma_{k-2} + \cdots + \phi_p \gamma_{k-p} \\ &\quad + \sigma_u^2 (\theta_k \psi_0 + \theta_{k+1} \psi_1 + \cdots)\end{aligned}$$

$$= \sum_{l=1}^p \phi_l \gamma_{k-l} + \sigma_u^2 \sum_{l=k}^q \theta_l \psi_{l-k} \quad (\theta_l = 0 \forall l \geq q+1)$$



For $k \geq q + 1$

$$\gamma_k = \phi_1 \gamma_{k-1} + \phi_2 \gamma_{k-2} + \cdots + \phi_p \gamma_{k-p}$$

$$= \sum_{l=1}^p \phi_l \gamma_{k-l}$$

Hence, for $k \geq q + 1$

$$\rho_k = \phi_1 \rho_{k-1} + \cdots + \phi_p \rho_{k-p}.$$

For $k \geq q + 1$ ACF behaves like that of an AR(p) process.



Topics

- Model selection: how do you choose between (and compare) alternate models?
- Multimodel inference: how do you combine information from more than 1 model?



Comparing alternative models

- We don't ask "Is the model right or wrong?" We ask "Do the data support one model more than a competing model?"
- Strength of evidence (support) for a model is relative:
 - Relative to other models: As models improve, support may change.
 - Relative to data at hand: As the data improve, support may change.



*But is likelihood enough?
The challenge of parsimony*

The importance of seeking simple answers...

“It will not be sufficient, when faced with a mass of observations, to plead **special creation**, even though, as we shall see, such a hypothesis commands a higher numerical likelihood than any other.”

(Edwards, 1992, pg. 1, in explaining the need for a rigorous basis for scientific inference, given uncertainty in nature...)



Models, Truth, and “Full Reality” (The Burnham and Anderson view...)

“We believe that “truth” (full reality) in the biological sciences has essentially infinite dimension, and hence ... cannot be revealed with only ... finite data and a “model” of those data...

... We can only hope to identify a model that provides a good *approximation* to the data available.”

(Burnham and Anderson 2002, pg. 20)



The “full” model

- Everything is the way it is because it is...
- In statistical terms, this is simply a model with as many parameters as observations
 - i.e.: $x_i = \theta_i$

*This will always be the model with the highest likelihood!
(but it won't be the most parsimonious)...*



Parsimony and Ockham's razor...

William of Ockham (1285-1349):

“Pluralitas non est ponenda sine neccesitate”
“entities should not be multiplied unnecessarily”



“Parsimony: ... **2** : economy in the use of means to an end;
especially : economy of explanation in conformity with Occam's
razor”

(*Merriam-Webster Online Dictionary*)



A more general framework for model comparison: Information theory

- “Reality” = “Truth” = Unknowable (or at least too much trouble to find...)
- Models are approximations of reality, and we’d like to know how “close” they are to reality...
- The “distance” between a model and reality is defined by the “Kullback-Leibler Information” (K-L distance)
- Unfortunately, K-L distance can only be directly computed in hypothetical cases where reality is known..

See Chapter 2 of Burnham and Anderson for discussion and details...



How much information is lost when using a simple model to approximate reality?

Answer: the Kullback-Leibler Distance (generally unknowable)

More Practical Answer: Akaike's Information Criterion (AIC) identifies the model that minimizes KL distance

$$AIC = -2\ln(L(\theta | x)) + 2K$$



Maximum Likelihood learning of Gaussians for Data Mining

- Why we should care
- Learning Univariate Gaussians
- Learning Multivariate Gaussians
- What's a biased estimator?
- Bayesian Learning of Gaussians



Why we should care

- Maximum Likelihood Estimation is a very very very very fundamental part of data analysis.
- “MLE for Gaussians” is training wheels for our future techniques
- Learning Gaussians is more useful than you might guess...



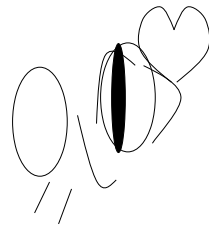
Learning Gaussians from Data

- Suppose you have $x_1, x_2, \dots, x_R \sim (\text{i.i.d}) N(\mu, \sigma^2)$
- But you don't know μ

(you do know σ^2)



MLE: For which μ is $x_1, x_2, \dots, x_R$ most likely?



MAP: Which μ maximizes $p(\mu | x_1, x_2, \dots, x_R, \sigma^2)$?

Despite this, we'll spend 95% of our time on MLE. Why? Wait and see...



Algebra Euphoria

$$\mu^{mle} = \arg \max_{\mu} p(x_1, x_2, \dots, x_R \mid \mu, \sigma^2)$$

$$= \arg \max_{\mu} \prod_{i=1}^R p(x_i \mid \mu, \sigma^2) \quad \text{(by i.i.d)}$$

$$= \arg \max_{\mu} \sum_{i=1}^R \log p(x_i \mid \mu, \sigma^2) \quad \text{(monotonicity of log)}$$

$$= \arg \max_{\mu} \frac{1}{\sqrt{2\pi} \sigma} \sum_{i=1}^R - \frac{(x_i - \mu)^2}{2\sigma^2} \quad \text{(plug in formula for Gaussian)}$$

$$= \arg \min_{\mu} \sum_{i=1}^R (x_i - \mu)^2 \quad \text{(after simplification)}$$



Intermission: A General Scalar MLE strategy

Task: Find MLE θ assuming known form for $p(\text{Data} \mid \theta, \text{stuff})$

1. Write $LL = \log P(\text{Data} \mid \theta, \text{stuff})$
2. Work out $\partial LL / \partial \theta$ using high-school calculus
3. Set $\partial LL / \partial \theta = 0$ for a maximum, creating an equation in terms of θ
4. Solve it*
5. Check that you've found a maximum rather than a minimum or saddle-point, and be careful if θ is constrained

*This is a perfect example of something that works perfectly in all textbook examples and usually involves surprising pain if you need it for something new.



The MLE μ

$$\mu^{mle} = \arg \max_{\mu} p(x_1, x_2, \dots, x_R \mid \mu, \sigma^2)$$

$$= \arg \min_{\mu} \sum_{i=1}^R (x_i - \mu)^2$$

$$= \mu \text{ s.t. } 0 = \frac{\partial \text{LL}}{\partial \mu} = \frac{\partial}{\partial \mu} \sum_{i=1}^R (x_i - \mu)^2$$
$$= - \sum_{i=1}^R 2(x_i - \mu)$$

$$\text{Thus } \mu = \frac{1}{R} \sum_{i=1}^R x_i$$



$$\mu^{mle} = \frac{1}{R} \sum_{i=1}^R x_i$$

- The best estimate of the mean of a distribution is the mean of the sample!



A General MLE strategy

Suppose $\theta = (\theta_1, \theta_2, \dots, \theta_n)^\top$ is a vector of parameters.

Task: Find MLE θ assuming known form for $p(\text{Data} \mid \theta, \text{stuff})$

1. Write $LL = \log P(\text{Data} \mid \theta, \text{stuff})$
2. Work out $\partial LL / \partial \theta$ using high-school calculus
3. Solve the set of simultaneous equations

$$\frac{\partial LL}{\partial \theta_1} = 0$$

$$\frac{\partial LL}{\partial \theta_2} = 0$$

$\vdots$

$$\frac{\partial LL}{\partial \theta_n} = 0$$

4. Check that you're at a maximum



MLE for univariate Gaussian

- Suppose you have $x_1, x_2, \dots, x_R \sim (\text{i.i.d}) N(\mu, \sigma^2)$
- But you don't know μ or σ^2
- MLE: For which $\theta = (\mu, \sigma^2)$ is $x_1, x_2, \dots, x_R$ most likely?

$$\log p(x_1, x_2, \dots, x_R \mid \mu, \sigma^2) = -R(\log \pi + \frac{1}{2} \log \sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^R (x_i - \mu)^2$$

$$\frac{\partial LL}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^R (x_i - \mu)$$

$$\frac{\partial LL}{\partial \sigma^2} = -\frac{R}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^R (x_i - \mu)^2$$



MLE for univariate Gaussian

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- But you don't know μ or σ^2
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$$\log p(x_1, x_2, \dots, x_R \mid \mu, \sigma^2) = -R(\log \pi + \frac{1}{2} \log \sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^R (x_i - \mu)^2$$

$$0 = \frac{1}{\sigma^2} \sum_{i=1}^R (x_i - \mu) \Rightarrow \mu = \frac{1}{R} \sum_{i=1}^R x_i$$

$$0 = -\frac{R}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^R (x_i - \mu)^2 \Rightarrow \text{what?}$$



MLE for univariate Gaussian

- Suppose you have $x_1, x_2, \dots, x_R \sim (\text{i.i.d}) N(\mu, \sigma^2)$
- But you don't know μ or σ^2
- MLE: For which $\theta = (\mu, \sigma^2)$ is $x_1, x_2, \dots, x_R$ most likely?

$$\mu^{mle} = \frac{1}{R} \sum_{i=1}^R x_i$$

$$\sigma_{mle}^2 = \frac{1}{R} \sum_{i=1}^R (x_i - \mu^{mle})^2$$



Unbiased Estimators

- An estimator of a parameter is **unbiased** if the expected value of the estimate is the **same** as the true value of the parameters.
- If $x_1, x_2, \dots, x_R \sim (\text{i.i.d}) N(\mu, \sigma^2)$ then

$$E[\mu^{mle}] = E\left[\frac{1}{R} \sum_{i=1}^R x_i\right] = \mu$$

μ^{mle} is unbiased



Biased Estimators

- An estimator of a parameter is **biased** if the expected value of the estimate is **different from** the true value of the parameters.
- If $x_1, x_2, \dots, x_R \sim (\text{i.i.d}) N(\mu, \sigma^2)$ then

$$E[\sigma_{mle}^2] = E\left[\frac{1}{R} \sum_{i=1}^R (x_i - \mu^{mle})^2\right] = E\left[\frac{1}{R} \left(\sum_{i=1}^R x_i - \frac{1}{R} \sum_{j=1}^R x_j\right)^2\right] \neq \sigma^2$$

σ_{mle}^2 is biased



MLE Variance Bias

- If $x_1, x_2, \dots, x_R \sim (\text{i.i.d}) N(\mu, \sigma^2)$ then

$$E[\sigma_{mle}^2] = E\left[\frac{1}{R} \left(\sum_{i=1}^R x_i - \frac{1}{R} \sum_{j=1}^R x_j \right)^2\right] = \left(1 - \frac{1}{R}\right) \sigma^2 \neq \sigma^2$$

Intuition check: consider the case of $R=1$

Why should our guts expect that σ_{mle}^2 would be an underestimate of true σ^2 ?

How could you prove that?



Unbiased estimate of Variance

- If $x_1, x_2, \dots, x_R \sim (\text{i.i.d}) N(\mu, \sigma^2)$ then

$$E[\sigma_{mle}^2] = E\left[\frac{1}{R} \left(\sum_{i=1}^R x_i - \frac{1}{R} \sum_{j=1}^R x_j \right)^2\right] = \left(1 - \frac{1}{R}\right) \sigma^2 \neq \sigma^2$$

So define $\sigma_{\text{unbiased}}^2 = \frac{\sigma_{mle}^2}{\left(1 - \frac{1}{R}\right)}$ So $E[\sigma_{\text{unbiased}}^2] = \sigma^2$



Unbiased estimate of Variance

- If $x_1, x_2, \dots, x_R \sim (\text{i.i.d}) N(\mu, \sigma^2)$ then

$$E[\sigma_{mle}^2] = E\left[\frac{1}{R} \left(\sum_{i=1}^R x_i - \frac{1}{R} \sum_{j=1}^R x_j \right)^2\right] = \left(1 - \frac{1}{R}\right) \sigma^2 \neq \sigma^2$$

So define $\sigma_{\text{unbiased}}^2 = \frac{\sigma_{mle}^2}{\left(1 - \frac{1}{R}\right)}$ So $E[\sigma_{\text{unbiased}}^2] = \sigma^2$

$$\sigma_{\text{unbiased}}^2 = \frac{1}{R-1} \sum_{i=1}^R (x_i - \mu^{mle})^2$$



Unbiasedity discussion

- *Which is best?*

$$\sigma_{mle}^2 = \frac{1}{R} \sum_{i=1}^R (x_i - \mu^{mle})^2$$

$$\sigma_{unbiased}^2 = \frac{1}{R-1} \sum_{i=1}^R (x_i - \mu^{mle})^2$$

Answer:

- It depends on the task
- And doesn't make much difference once $R \rightarrow \text{large}$



MLE for m-dimensional Gaussian

- Suppose you have $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_R \sim (\text{i.i.d}) \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$
- But you don't know $\boldsymbol{\mu}$ or $\boldsymbol{\Sigma}$
- MLE: For which $\boldsymbol{\theta} = (\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_R$ most likely?

$$\boldsymbol{\mu}^{mle} = \frac{1}{R} \sum_{k=1}^R \mathbf{x}_k$$

$$\boldsymbol{\Sigma}^{mle} = \frac{1}{R} \sum_{k=1}^R (\mathbf{x}_k - \boldsymbol{\mu}^{mle})(\mathbf{x}_k - \boldsymbol{\mu}^{mle})^T$$



MLE for m-dimensional Gaussian

- Suppose you have $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_R \sim (\text{i.i.d}) \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$
- But you don't know $\boldsymbol{\mu}$ or $\boldsymbol{\Sigma}$
- MLE: For which $\boldsymbol{\theta} = (\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_R$ most likely?

$$\boldsymbol{\mu}^{mle} = \frac{1}{R} \sum_{k=1}^R \mathbf{x}_k$$

$$\mu_i^{mle} = \frac{1}{R} \sum_{k=1}^R x_{ki}$$

$$\boldsymbol{\Sigma}^{mle} = \frac{1}{R} \sum_{k=1}^R (\mathbf{x}_k - \boldsymbol{\mu}^{mle})(\mathbf{x}_k - \boldsymbol{\mu}^{mle})^T$$

Where $1 \leq i \leq m$

And x_{ki} is value of the i^{th} component of $\mathbf{x}_k$ (the i^{th} attribute of the k^{th} record)

And μ_i^{mle} is the i^{th} component of $\boldsymbol{\mu}^{mle}$



MLE for m-dimensional Gaussian

- Suppose you have $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_R \sim (\text{i.i.d}) N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$
- But you don't know $\boldsymbol{\mu}$ or $\boldsymbol{\Sigma}$
- MLE: For which $\boldsymbol{\theta} = (\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_R$ most likely?

$$\boldsymbol{\mu}^{mle} = \frac{1}{R} \sum_{k=1}^R \mathbf{x}_k$$

$$\boldsymbol{\Sigma}^{mle} = \frac{1}{R} \sum_{k=1}^R (\mathbf{x}_k - \boldsymbol{\mu}^{mle})(\mathbf{x}_k - \boldsymbol{\mu}^{mle})^T$$

Where $1 \leq i \leq m, 1 \leq j \leq m$

And x_{ki} is value of the i^{th} component of $\mathbf{x}_k$ (the i^{th} attribute of the k^{th} record)

And σ_{ij}^{mle} is the $(i,j)^{\text{th}}$ component of $\boldsymbol{\Sigma}^{mle}$

$$\sigma_{ij}^{mle} = \frac{1}{R} \sum_{k=1}^R (x_{ki} - \mu_i^{mle})(x_{kj} - \mu_j^{mle})$$



MLE for m-dimensional Gaussian

- Suppose you have $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_R \sim$
- But you don't know μ or Σ
- MLE: For which $\theta = (\mu, \Sigma)$ is $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_R$

$$\mu^{mle} = \frac{1}{R} \sum_{k=1}^R \mathbf{x}_k$$

$$\Sigma^{mle} = \frac{1}{R} \sum_{k=1}^R (\mathbf{x}_k - \mu^{mle})(\mathbf{x}_k - \mu^{mle})^T$$

$$\Sigma^{\text{unbiased}} = \frac{\Sigma^{mle}}{1 - \frac{1}{R}} = \frac{1}{R-1} \sum_{k=1}^R (\mathbf{x}_k - \mu^{mle})(\mathbf{x}_k - \mu^{mle})^T$$

Q: How would you prove this?

A: Just plug through the MLE recipe.

Note how Σ^{mle} is forced to be symmetric non-negative definite

Note the unbiased case

How many datapoints would you need before the Gaussian has a chance of being non-degenerate?



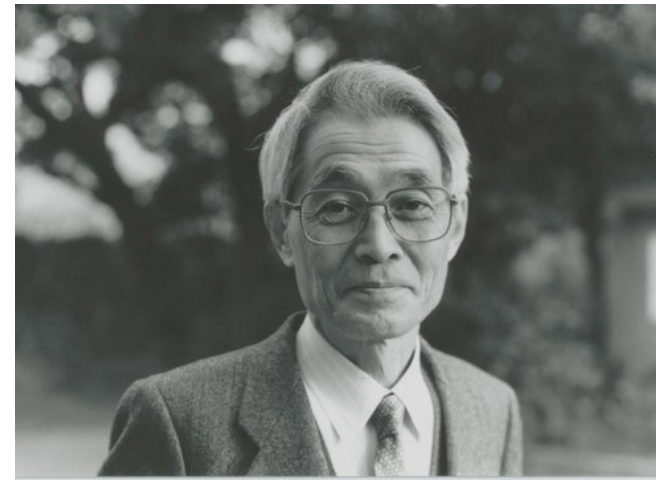
Akaike's contribution (1973)

- Akaike (1973) proposed “an information criterion” (AIC) (but now often called an Akaike Information Criterion) that relates likelihood to K-L distance, and includes an explicit term for model complexity...

$$AIC = -2 \ln(L(\hat{\theta} | y)) + 2K$$

This is an estimate of ***the expected, relative distance between the fitted model and the unknown true mechanism that generated the observed data.***

K =number of estimated parameters





General guidelines for use of AIC

- We select the model with **smallest** value of AIC (i.e. closest to “truth”).
- AIC will identify the best model in the set, even if all the models are poor!
- It is the researcher’s (your) responsibility that the set of candidate models includes well founded, realistic models.



AIC for small samples

- Unless the sample size (n) is large with respect to the number of estimated parameters (K), use of AIC_c is recommended.

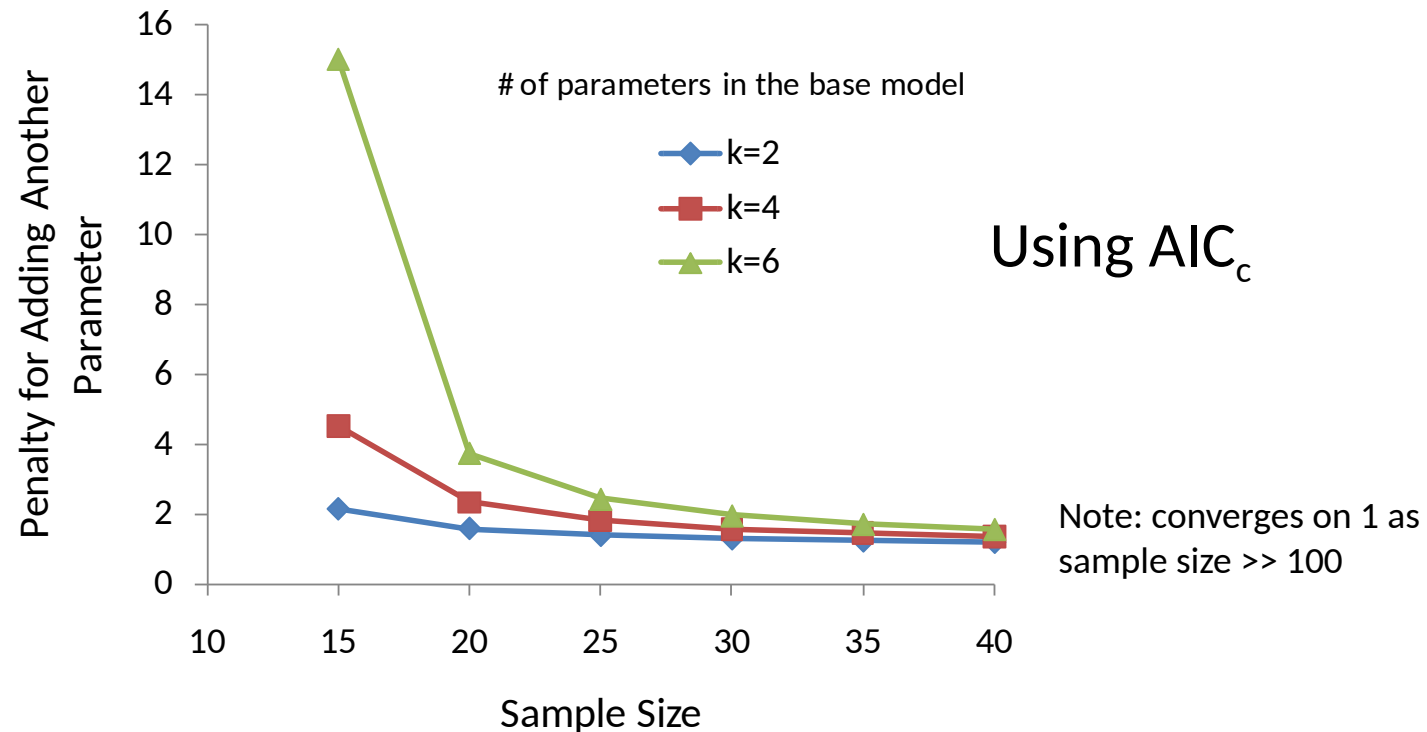
$$AIC_c = -2 \ln(L(\theta | y)) + 2K \left(\frac{n}{n - K - 1} \right)$$

- Generally, you should use AIC_c when the ratio of n/K is small (less than ~ 40), based on K from the global (most complicated) model.
- Use AIC or AIC_c consistently in an analysis rather than mix the two criteria.



Penalty for adding parameters - small sample sizes -

How much does likelihood have to improve to justify adding one more parameter – as a function of sample size and model complexity?





But what about cases with very large sample sizes?

- Does it make intuitive sense that a model that is only 1 unit of likelihood better, for a sample with 10000 observations, should be selected?
- The Schwarz criterion (also known as the Bayesian Information Criterion)

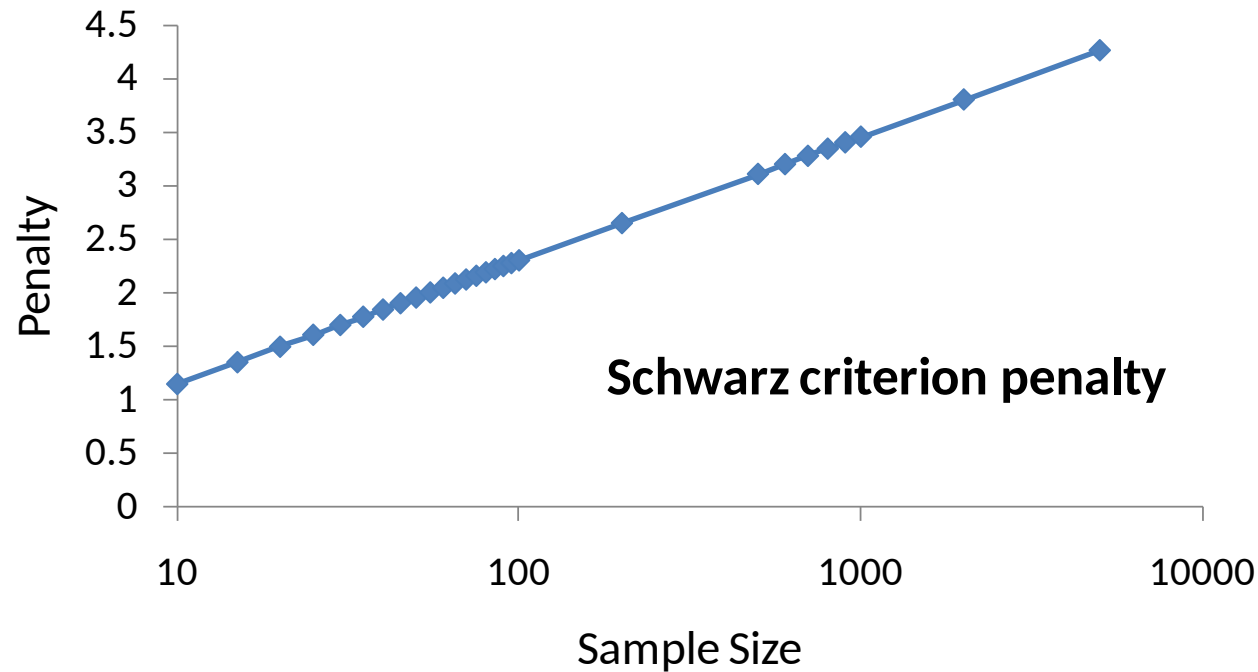
$$SC = -2 \ln(L(\theta | y)) + k \ln(n)$$

So the penalty is simply a function of sample size, and increases as a function of the natural log of n (divided by 2)



Penalty for adding parameters - large sample sizes -

$$SC = -2\ln(L(\theta | y)) + k\ln(n)$$



So the penalty is simply a function of sample size (regardless of base model complexity), and increases as a function of the natural log of n (divided by 2)



Some Rough Rules of Thumb

- Differences in AIC (Δ_i 's) can be used to interpret strength of evidence for one model vs. another.
- A model with a Δ value within 1-2 of the best model has substantial support in the data, and should be considered along with the best model.
- A Δ value within only 4-7 units of the best model has considerably less support.
- A Δ value > 10 indicates that the worse model has virtually no support and can be omitted from further consideration.



Strength of evidence for alternate models: Akaike weights

Akaike weights (w_i) are ***the weight of evidence in favor of model i being the actual best model*** for the situation at hand given that one of the N models must be the best model ***for that set of N models.***

$$w_i = \frac{\exp(-0.5\Delta_i)}{\sum_{r=1}^N \exp(-0.5\Delta_r)} \quad \text{where} \quad \Delta_i = AIC_i - AIC_{min}$$

Akaike weights for all models combined should add up to 1.



Uses of Akaike weights

- “Probability” that the candidate model is the best model.
- Relative strength of evidence (evidence ratios).
- Variable selection—which independent variable has the greatest influence?
- Model averaging.



An example...

The Data:

x_i = measurements of DBH on 50 trees

y_i = measurements of crown radius on those trees

The Scientific Models:

$$y_i = \beta x_i + \varepsilon \quad [1 \text{ parameter } (\beta)]$$

$$y_i = \alpha + \beta x_i + \varepsilon \quad [2 \text{ parameters } (\alpha, \beta)]$$

$$y_i = \alpha + \beta x_i + \gamma x_i^2 + \varepsilon \quad [3 \text{ parameters } (\alpha, \beta, \gamma)]$$

The Probability Model:

ε is normally distributed, with mean = 0 and variance estimated from the observed variance of the residuals...



Back to the example.....

						Likelihood	No. parameters
Model 1: Radius =	$B \cdot \text{DBH}$					130.42	1
Model 2: Radius =	$A + B \cdot \text{DBH}$					108.01	2
Model 3: Radius =	$A + B \cdot \text{DBH} + C \cdot (\text{DBH}^2)$					107.99	3

						AIC	Akaike weight
Model 1: Radius =	$B \cdot \text{DBH}$					264.84	0.00
Model 2: Radius =	$A + B \cdot \text{DBH}$					222.02	0.73
Model 3: Radius =	$A + B \cdot \text{DBH} + C \cdot (\text{DBH}^2)$					223.97	0.27

Akaike weights can be interpreted as the estimated probability that model i is the best model for the data at hand, given the set of models considered. Weights > 0.90 indicate that robust inferences can be made using just that model.



Ambivalence about selecting a best model to use for inference...

The inability to identify a single best model is not a defect of the AIC method. It is an indication that the ***data are not adequate to reach strong inference.***

What is to be done??

MULTIMODEL INFERENCE AND MODEL
AVERAGING



Summary: Steps in Model Selection

- Develop candidate models based on biological knowledge.
- Take observations (data) relevant to predictions of the model.
- Use data to obtain MLE of parameters of the alternate models.
- Evaluate strength of evidence for alternate models using AIC and Akaike weights.
- ...Multimodel Inference?



Bias and Uncertainty in Model Selection

- Model Selection Bias: Chance inclusion of meaningless variables in a model will produce a biased underestimate of the variance, and a corresponding exaggeration of the precision of the model
- Model Selection Uncertainty: The fact that we are using data (with uncertainty) to both estimate parameters and to select the best model necessarily introduces uncertainty into the model selection process